

3-Mark Questions & Answers — DBMS Module 2: Relational Algebra & Calculus

Q1. Explain Set Difference (−) and Intersection (∩) operations with examples.

1. **Set Difference ($R - S$):** Returns tuples that belong to relation R but NOT to relation S (both must be union-compatible).
2. **Intersection ($R \cap S$):** Returns tuples present in BOTH relations R and S.
- Identity: $R \cap S = R - (R - S)$

Q2. Explain Natural Join ($\bowtie$) vs Theta Join ($\bowtie_{\theta}$).

Feature	Natural Join ($\bowtie$)	Theta Join ($\bowtie_{\theta}$)
Join Condition	Equality on ALL common attribute names automatically.	Explicit conditional predicate (e.g., $R.A > S.B$ or $R.A = S.B$).
Duplicate Columns	Eliminates duplicate common columns in the result.	Retains columns from both relations unless projected out.

Q3. Differentiate between Procedural and Non-Procedural query languages.

Feature	Relational Algebra (Procedural)	Relational Calculus (Non-Procedural)
Approach	Tells HOW to retrieve data step-by-step.	Tells WHAT data to retrieve without retrieval steps.
Execution	Specifies exact sequence of relational operations.	Uses first-order predicate calculus logic formulas.
Foundation	Basis for DBMS physical query execution plans.	Theoretical foundation for SQL declarative language.